

Gradient Descent Manual Calculation.

1. Computing the predicted values $\hat{y}$ for each data point

$$\text{Model: } \hat{y} = (m_1 \times x_1) + (m_2 \times x_2) + b$$

x_1 and x_2 are input features
 m_1, m_2 and b are parameters.

For Sample 1: $x_1 = 1, x_2 = 3$ and $y = 5$
Sample 2: $x_1 = 4, x_2 = 10$ and $y = 6$
 $m_1 = -1, m_2 = 2.$

$$\begin{aligned}\hat{y}_1 &= (m_1 \times x_1) + (m_2 \times x_2) + b \\ &= (-1 \times 1) + (2 \times 3) + 1 \\ &= (-1 \times 6) + 1 \\ &= -6 + 1 \\ \hat{y}_1 &= -5\end{aligned}$$

$$\begin{aligned}\hat{y}_2 &= (-1 \times 4) + (2 \times 10) + 1 \\ &= (-4 \times 20) + 1 \\ &= -80 + 1 \\ \hat{y}_2 &= -79\end{aligned}$$

$$\begin{aligned}e_1 &= \hat{y}_1 - y_1 \\ &= -5 - 5 \\ e_1 &= -10\end{aligned}$$

$$\begin{aligned}e_2 &= \hat{y}_2 - y_2 \\ &= -79 - 6 \\ e_2 &= -85\end{aligned}$$

$$\therefore \underline{\underline{e = [-10, -85]}}$$

2. Derive the gradient of MSE Cost Function $J(m, b)$.

$$J(m, b) = \frac{1}{n} \sum (\hat{y} - y)^2 \quad \text{But } n=2$$

$$= \frac{1}{2} \sum (\hat{y} - y)^2$$

$$= \frac{1}{2} \cdot (\hat{y} - y)^2$$

$$\text{but } e = \hat{y} - y$$

Therefore;

$$J = \frac{1}{2}$$

Therefore, replace $\hat{y} - y$ with e .

$$J = \frac{1}{2} \sum e^2$$

But we have 2 number of samples. Therefore;

$$J = \left(\frac{1}{2}\right) \times (e_1^2 + e_2^2)$$

$$\text{But } e_1 = m_1 x_{11} + m_2 x_{21} + b_1 - y_1$$

$$e_2 = m_1 x_{12} + m_2 x_{22} + b_2 - y_2$$

$$\begin{aligned} \text{Using Our data, } e_1 &= m_1 \cdot 1 + m_2 \cdot 3 + b_1 - 5 \\ &= m_1 + 3m_2 + b_1 - 5 \\ &= 3m_1 + m_2 + b_1 - 5 \end{aligned}$$

$$\begin{aligned} e_2 &= m_1 \cdot 4 + m_2 \cdot 10 + b_2 - 6 \\ &= 4m_1 + 10m_2 + b_2 - 6 \\ &= 40m_1 + m_2 + b_2 - 6 \end{aligned}$$

$$\text{Compute } dJ/dm_1 = \frac{dJ}{de_1} \times \frac{de_1}{dm_1} + \frac{dJ}{de_2} \times \frac{de_2}{dm_1}$$

$$\text{Computed } \frac{dJ}{de_1} : J = \frac{1}{2} (e_1^2 + e_2^2)$$

$$\frac{dJ}{de_1} = \frac{1}{2} \times 2e_1$$

$$\frac{dJ}{de_1} = e_1$$

Compute $\frac{de_1}{dm_1}$: $e_1 = 3m_1 m_2 + b_1 - 5$

$\frac{de_1}{dm_1} = 3m_2 \rightarrow$ Only m_1 varies, everything else is constant

Compute $\frac{dJ}{de_2}$: $\frac{dJ}{de_2} = \frac{1}{2} \cdot 2e_2$
 $\frac{dJ}{de_2} = e_2$

Compute $\frac{de_2}{dm_1}$: $e_2 = 40m_1 m_2 + b_2 - 6$
 $\frac{de_2}{dm_1} = 40m_2$

Combine using chain rule: $\frac{dJ}{dm_1} = e_1 \times 3m_2 + e_2 \times 40m_2$
 $= m_2 (3e_1 + 40e_2)$

$\therefore \frac{dJ}{dm_1} = m_2 (3e_1 + 40e_2)$

For dJ/db_2 :

Compute $\frac{de_2}{db_2}$: $e_2 = 40m_1 m_2 + b_2 - 6$
 $\frac{de_2}{db_2} = 1$

Combine using chain rule

$\frac{dJ}{db_2} = \frac{dJ}{de_2} \times \frac{de_2}{db_2}$
 $= e_2 \times 1$

$\frac{dJ}{db_2} = e_2$

For dJ/dm_2

Compute $\frac{de_1}{dm_2}$: $e_1 = 3m_1 m_2 + b_1 - 5$

$$\frac{de_1}{dm_2} = 3m_1$$

Compute $\frac{de_2}{dm_2}$: $e_2 = 40m_1 m_2 + b_2 - 6$

$$\frac{de_2}{dm_2} = 40m_1$$

Combine using chain rule

$$\frac{dJ}{dm_2} = \frac{dJ}{de_1} \times \frac{de_1}{dm_2} + \frac{dJ}{de_2} \times \frac{de_2}{dm_2}$$

$$= e_1 \times 3m_1 + e_2 \times 40m_1$$

$$= m_1 (3e_1 + 40e_2)$$

$$\therefore \frac{dJ}{dm_2} = m_1 (3e_1 + 40e_2)$$

For dJ/db_1

Compute $\frac{de_1}{db_1}$: $e_1 = 3m_1 m_2 + b_1 - 5$

$$\frac{de_1}{db_1} = 1$$

Combine using chain rule:

$$\frac{dJ}{db_1} = \frac{dJ}{de_1} \times \frac{de_1}{db_1}$$

$$= e_1 \times 1$$

$$\therefore \frac{dJ}{db_1} = e_1$$

ITERATION 0

Starting: $m = [-1, 2]$ $b = [1, 1]$ $x = [1, 3]$ $y = 5$
 $x = [4, 10]$ $y = 6$

Predictions:

$$\hat{y}_1 = m_1 x_1 + m_2 x_2 + b_1$$

$$= -1 \times 1 + 2 \times 3 + 1$$

$$= -1 \times 6 + 1$$

$$\hat{y}_1 = -5$$

$$\hat{y}_2 = -1 \times 4 + 2 \times 10 + 1$$

$$= -4 \times 20 + 1$$

$$\hat{y}_2 = -79$$

Compute Errors ($e = \hat{y} - y$)

$$e_1 = \hat{y}_1 - y_1 = -5 - 5 = -10$$

$$e_2 = \hat{y}_2 - y_2 = -79 - 6 = -85$$

Compute Gradient: $\frac{dJ}{dm_1} = m_2 (3e_1 + 40e_2)$

$$= 2 (3 \times -10 + 40 \times -85)$$

$$dJ/dm_1 = -6860$$

$$\frac{dJ}{dm_2} = m_1 (3e_1 + 40e_2)$$

$$= -1 (3 \times -10 + 40 \times -85)$$

$$dJ/dm_2 = 3430$$

$$\frac{dJ}{db_1} = e_1 = -10$$

$$\frac{dJ}{db_2} = e_2 = -85$$

Update Parameters: $\alpha = 0.00001$ (Because gradient magnitude is large)
New parameter = Old parameter - $\alpha \frac{dJ}{d\text{parameter}}$

$$\text{For } m_1 = -1 - 0.00001(-6860) \\ = -0.9314$$

$$m_2 = 2 - 0.00001(3430) \\ = 1.9657$$

$$b_1 = 1 - 0.00001(-10) \\ = 1.0001$$

$$b_2 = 1 - 0.00001(-85) \\ = 1.00085$$

Updated state after iteration 0:

$$m = [-0.9314, 1.9657]$$

$$b = [1.0001, 1.00085]$$

ITERATION 1

$$\text{Starting state: } m = [-0.9314, 1.9657] \\ b = [1.0001, 1.00085]$$

Compute predictions:

$$\hat{y}_1 = m_1 x_1 + m_2 x_2 + b_1$$

$$= (-0.9314 \times 1) + (1.9657 \times 3) + 1.0001$$

$$= -4.492458940000001$$

$$\hat{y}_2 = (-0.9314 \times 1.9657) \times (4 \times 16) + 1.00085$$

$$= -72.2332692$$

Compute errors:

$$e_1 = -4.492458940000001 - 5 = -9.49245894$$

$$e_2 = -72.2332692 - 6 = -78.2332692$$

Compute Gradient

$$\frac{dJ}{dm_1} = m_2(3e_1 + 40e_2) = 1.9657 \times (3 \times -9.49245894 + 40 \times -78.2332692)$$

$$= -6207.303470272674$$

$$\frac{dJ}{dm_2} = m_1(3e_1 + 40e_2)$$

$$= -0.9314 \times 3157.8081448199996$$

$$dJ/dm_2 = 2941.1825060853475$$

$$dJ/db_1 = e_1 = -9.49245894$$

$$dJ/db_2 = e_2 = -78.2332692$$

Update parameters $\alpha = 0.00001$

$$m_1 = -0.9314 - 0.00001 \times (-6207.303470272674)$$

$$m_1 = -0.8693269652972733$$

$$m_2 = 1.9657 - 0.00001(2941.1825060853475)$$

$$= 1.9362881749391465$$

$$b_1 = 1.0001 - 0.00001(-9.49245894) = 1.0001949245894$$

$$b_2 = 1.00085 - 0.00001(-78.2332692) = 1.001632332692$$

Updated state after Iteration 1

$$m_2 = [-0.8693269652972733, 1.9362881749391465]$$

$$b_2 = [1.0001949245894, 1.001632332692]$$

ITERATION 2

Starting state

$$m = [-0.8693269652972733, 1.9362881749391465]$$

$$b = [1.0001949245894, 1.001632332692]$$

Compute prediction:

$$\begin{aligned}\hat{y}_1 &= m_1 m_2 x_1 x_2 + b \\ &= (-0.8693269652972733 \times 1.9362881749391465) \times (1 \times 3) + \\ &\quad 1.0001949245894\end{aligned}$$

$$\hat{y}_1 = -4.049607644593132$$

$$\begin{aligned}\hat{y}_2 &= (-0.8693269652972733 \times 1.9362881749391465) (4 \times 10) + \\ &\quad 1.001632332692\end{aligned}$$

$$\hat{y}_2 = -66.32906858974177$$

Compute Error:

$$e_1 = -4.049607644593132 - 5 = -9.04960764459313$$

$$e_2 = -66.32906858974177 - 6 = -72.32906858974177$$

Compute Gradient

$$dJ/dm_1 = m_2(3e_1 + 40e_2)$$

$$= 1.9362881749391465(3x - 9.04960764459313 + 40x \\ 72.32906858974177)$$

$$dJ/dm_1 = -5654.56475397371$$

$$dJ/dm_2 = 0.8698269652972733x - 2920.31156652345$$

$$= 2538.705591848357$$

$$dJ/db_1 = e_1 = -9.04960764459313$$

$$dJ/db_2 = e_2 = -72.32906858974177$$

Update parameters

$$m_1 = -0.8698269652972733 - 0.0001(-5654.56475397371)$$

$$= -0.8698269652972733 + 0.565456475397371$$

$$m_2 = 1.9362881749391465 - 0.0001(2538.705591848357)$$

$$= 1.9109011190206628$$

$$b_1 = 1.000949245894 - 0.0001(-9.04960764459313)$$

$$= 1.000949245894 + 0.000904960764459313$$

$$b_2 = 1.00162332692 - 0.0001(-72.32906858974177)$$

$$= 1.00162332692 + 0.0007232906858974177$$

Updated state after iteration 2.

$$m_3 = [-0.8127813177632995, 1.9109011190206628]$$

$$b_3 = [1.000949245894, 1.0023556233778974]$$

~~ITERATION 3~~ ITERATION 3.

Starting state

$$m = [-0.8127813177632995, 1.910901119020628]$$

$$b = [1.0002854206658458, 1.0023556233778974]$$

Compute prediction

$$\hat{y}_1 = (-0.813 \times 1.911) \times (1 \times 3) + 1.0003 = -3.66$$

$$\hat{y}_2 = (-0.813 \times 1.911) \times (4 \times 10) + 1.0024 = -61.12$$

Compute error

$$e_1 = -3.66 - 5 = -8.66$$

$$e_2 = -61.12 - 6 = -67.12$$

Compute Gradient

$$\begin{aligned} dJ/dm_1 &= 1.911 \times (3 \times -8.66) + 40 \times -67.12 \\ &= -5180 \end{aligned}$$

$$\begin{aligned} dJ/dm_2 &= -0.813 \times -2711 \\ &= 2203 \end{aligned}$$

$$dJ/db_1 = e_1 = -8.66$$

$$dJ/db_2 = -67.12$$

Update parameter. $\lambda = 0.00001$

$$m_1 = -0.812 - 0.00001(-5180)$$

$$= -0.761$$

$$m_2 = 1.911 - 0.00001(2203)$$

$$= 1.889$$

$$mb_1 = 1.0003 - 0.00001(-8.66)$$

$$= 1.0004$$

$$b_2 = 1.0024 - 0.00001(-67.12)$$

$$= 1.0031$$

$$\text{Final state : } m_4 = [-0.761, 1.889]$$

$$b_4 = [1.0004, 1.0031]$$

7 Describing the trend

Iteration	e_1	e_2	m_1	m_2	b_1	b_2
0 (Start)	-10	-85	-1	2	1	1
1	-9.49	-78.23	-0.9814	1.9657	1.0001	1.00085
2	-9.05	-72.33	-0.8693	1.9363	1.0002	1.0016
3	-8.66	-67.12	-0.8128 -0.761	1.9109	1.0003	1.0024
4 (Final)	-0.761	-68.89	-0.761	1.889	1.0004	1.0031

Computing the cost function $J = \frac{1}{2} (e_1^2 + e_2^2)$ at the start and end confirms this trend numerically:

$$J_0 = \frac{1}{2} (-10^2 + -85^2) = \frac{1}{2} (100 + 7225)$$

$$J_0 = 3662.5$$

$$J_3 = \frac{1}{2} (-8.66^2 + -67.12^2) = \frac{1}{2} (75 + 4505.1)$$

$$J_3 = 2290.05$$

Conclusion: Parameters are moving to reduce error, all parameters values are moving in directions that reduce prediction error.

Evidence:- errors e_1 and e_2 are decreasing in magnitude

- The cost J is also decreasing confirming that the gradient descent algorithm is moving toward lower error
- Parameters are adjusting systematically

Therefore: This confirms that the gradient descent algorithm is working correctly and the use of a small learning rate was ~~to~~ because of the large magnitudes.